

KAKINADA, India

WMO: 431890

Lat: 16.950N

Lon: 82.233E

Elev: 8

StdP: 101.23

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.1	19.9	14.0	9.9	24.7	15.2	10.8	24.1	6.4	26.3	4.7	26.0	0.2	320	0.320

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	8.6	39.6	27.4	37.7	27.6	36.4	27.5	29.5	34.7	29.0	34.2	28.6	33.5	1.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.2	24.5	32.2	27.8	23.9	31.8	27.5	23.4	31.5	97.4	35.3	95.2	34.2	93.2	33.6	33.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.3	4.0	3.3	DB	16.9	43.6	1.1	2.2	16.2	45.2	15.5	46.5	14.9	47.7	14.2	49.3	
			WB	14.8	30.5	1.1	1.2	14.0	31.4	13.3	32.1	12.7	32.8	11.8	33.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	24.8	26.5	28.9	31.1	32.6	31.4	29.5	29.2	29.2	28.5	26.6	25.0	(6)
	DBStd	2.88	1.18	1.24	1.28	1.36	2.09	2.60	1.97	1.60	1.48	1.38	1.24	1.21	(7)
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD10.0	6794	459	462	586	632	702	641	604	596	576	575	498	464	(10)
	CDD18.3	3753	201	228	328	382	444	391	346	338	326	317	248	206	(11)
	CDH23.3	44432	1410	2068	3809	5109	6446	5446	4376	4178	3966	3688	2417	1520	(12)
	CDH26.7	20575	314	719	1703	2766	3987	3105	2011	1808	1671	1443	718	331	(13)
Wind	WSAvg	1.0	0.8	0.8	0.8	1.0	1.1	1.3	1.5	1.3	0.9	0.8	1.0	0.9	(14)
Precipitation	PrecAvg	1048	6	6	10	21	31	115	173	154	172	217	124	17	(15)
	PrecMax	1491	42	30	166	154	224	226	299	375	479	565	654	141	(16)
	PrecMin	630	0	0	0	0	0	11	79	17	31	40	0	0	(17)
	PrecStd	236	11	10	34	33	53	59	64	95	99	149	158	31	(18)
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.0	34.1	35.7	37.9	43.4	42.5	37.7	35.6	35.5	35.0	32.4	30.9	(19)
		MCWB	23.8	23.2	25.8	27.7	25.8	26.7	26.5	27.2	27.2	26.6	24.6	23.6	(20)
	2%	DB	30.0	32.5	34.6	36.7	40.3	39.5	35.7	34.3	34.3	33.7	31.5	30.0	(21)
		MCWB	23.3	24.0	25.8	27.5	27.3	27.3	26.7	27.3	27.2	26.6	24.9	23.6	(22)
	5%	DB	29.2	31.4	33.7	35.8	38.2	37.4	34.3	33.4	33.4	32.8	30.8	29.3	(23)
		MCWB	23.0	24.1	25.7	27.4	27.9	27.4	26.9	27.2	27.3	26.4	24.7	23.1	(24)
	10%	DB	28.5	30.4	32.9	34.9	36.7	35.6	33.1	32.5	32.4	31.9	30.0	28.6	(25)
		MCWB	22.7	23.7	25.6	27.1	28.1	27.2	26.9	27.0	27.1	26.2	24.3	22.6	(26)
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.3	26.3	27.9	29.2	30.3	29.9	28.7	28.7	28.6	28.4	27.2	25.5	(27)
		MCDB	29.3	31.5	33.4	34.8	35.9	35.3	33.5	32.8	32.8	32.7	30.5	29.2	(28)
	2%	WB	24.5	25.5	27.1	28.6	29.6	29.2	28.1	28.1	28.2	27.7	26.5	24.8	(29)
		MCDB	28.4	30.5	32.6	34.1	34.9	34.4	32.5	32.1	32.3	31.8	29.7	28.4	(30)
	5%	WB	23.8	24.9	26.7	28.2	29.3	28.7	27.7	27.7	27.7	27.3	25.9	24.2	(31)
		MCDB	27.7	29.7	31.9	33.2	34.4	33.7	31.8	31.5	31.6	31.1	29.1	27.7	(32)
	10%	WB	23.3	24.4	26.2	27.7	28.9	28.2	27.3	27.3	27.4	26.9	25.4	23.7	(33)
		MCDB	27.2	28.8	31.1	32.6	33.9	32.9	31.2	31.0	31.0	30.4	28.5	27.1	(34)
Mean Daily Temperature Range		MDBR	8.3	8.8	8.9	8.7	8.6	7.4	5.8	5.6	5.8	6.3	7.3	8.0	(35)
	5% DB	MCDBR	8.9	10.1	9.7	9.5	10.8	10.5	7.9	7.2	7.5	8.0	8.1	8.5	(36)
		MCWBR	3.8	3.8	3.1	2.7	3.1	2.9	2.4	2.4	2.6	3.0	3.4	3.8	(37)
	5% WB	MCDBR	7.9	8.6	8.7	8.7	8.8	8.5	6.7	6.4	6.4	6.6	6.3	7.4	(38)
		MCWBR	3.8	3.6	3.2	3.1	3.6	3.4	2.7	2.7	2.8	3.0	3.1	3.8	(39)
Clear-Sky Solar Irradiance	taub	0.514	0.506	0.575	0.584	0.643	0.633	0.578	0.558	0.555	0.552	0.526	0.540	(40)	
	taud	1.990	2.034	1.889	1.917	1.792	1.822	1.968	2.027	2.034	2.023	2.024	1.945	(41)	
	Ebn at Noon	769	802	760	754	701	701	742	764	765	754	753	729	(42)	
	Edh at Noon	171	171	204	199	222	213	185	176	174	171	164	175	(43)	
All-Sky Solar Radiation	RadAvg	4.79	5.62	6.16	6.51	6.05	4.66	4.22	4.29	4.49	4.61	4.52	4.36	(44)	
	RadStd	0.18	0.21	0.25	0.25	0.23	0.45	0.52	0.41	0.38	0.47	0.36	0.26	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	+0.53	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+0.30	+0.30	N/A	N/A	+0.20	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page